

3.1 Current event curation

We consider four primary event categories for MultiVENT: Disasters, political events, social events, and technology events. We include thirteen current events per category for each target language. We use Google Trends statistics to select these events, based on its tracking of term popularity based on internet activity by country. We construct lists of the top five countries with the most speakers of each target language and review the top trending topics on Google in each of these countries over the last ten years. We record topics and search phrases that corresponded to current events that (1) align with one of the predefined event categories and (2) have sufficient online video coverage. For categories that did not amass a sufficient list of current events per language through this process, we consult Wikipedia’s yearly summaries of events to fill the remaining slots. Detailed statistics characterizing this set of current events are shown in Figure 2. As shown, the majority of selected events take place in the last few years, with only three taking place before 2016.

Also shown in Figure 2, there is not a bijective mapping between the language used in event coverage and the country the event took place in. The language and country are often related, e.g., Russian news content in MultiVENT predominantly takes place in Russia, but this is not true of all events in the dataset. For example, we include data in Chinese pertaining to the 2023 ATP tennis circuit in Dallas, Texas: At this event, tennis player Wu Yibing became the highest-ranked Chinese player in the history of the ATP rankings, and so the event received substantial Chinese news coverage. In cases such as this, news in multiple languages will heavily focus on the same current event, such as sports events and international political relations. We do not include the same event in multiple languages in MultiVENT by design, in contrast with data collection procedures used for efforts such as AIDA [44] which aim to cover a small collection of current events in many languages.

Every current event in the dataset is grounded in an English natural language document and, if the event is tagged with a non-English language, an additional natural language document in that target language. First, we check if a full English Wikipedia article exists for the current event. If not, we manually find a Wikipedia article that includes a passage describing the event. If Wikipedia does not have a passage that appropriately grounds the event, then a news article in English is selected as a grounding document instead. This process is then repeated for the target language. The dataset includes 468 grounding articles in total: 313 are full Wikipedia articles, 104 are Wikipedia passages, and 51 are external articles.

3.2 Video collection

We aim to collect visually and semantically distinct videos for each current event with an even split between firsthand witness accounts (e.g., first-person smartphone videos), amateur edited videos (e.g., vlogs), and professional news reports and compilations. Information regarding the resultant distribution of these categories and their semantic differences is included in Section 4.2. For each current event, we collect ten videos in the current event’s target language. We search YouTube and Twitter for these videos using target keywords collected from the Google Trends search and Wikipedia. After collecting the videos, we manually identify and remove duplicates, resulting in 2,396 videos in total. We do not include repeat videos, but sometimes professional news reports include firsthand footage that is already included as unedited footage in the dataset. In these cases, we keep both the news report and the original footage as the context and text metadata between the two are distinct. If the video has a natural language description, we tag the video with this description. If it does not, we use the video title as the tagged natural language description. We report the distribution of videos by source in Figure 2.

4 Data analysis

We present an analysis of MultiVENT to help characterize how online multimodal content contributes to our understanding of current events. We explore multimodal event coverage from three angles: (1) what kinds of information news videos contribute, (2) the differences in informative content provided by different types of news videos, and (3) how multimodal coverage of an event can evolve over time.

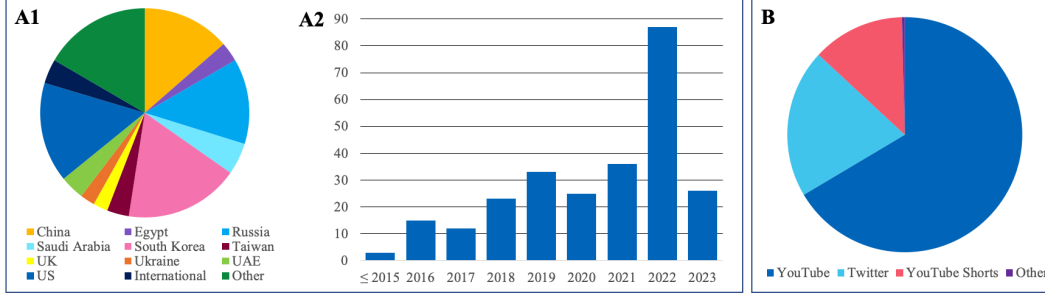


Figure 2: **A**: Statistics illustrating the distribution of current events selected for the dataset. (**A1**) depicts the general breakdown of countries in which each current event takes place. Many countries had a single current event, particularly small countries in the middle east and southeast Asia, and are consolidated into the "other" category for easier graph interpretation. (**A2**) shows the distribution of years during which the current events take place. **B**: Breakdown of data sources for the videos in the dataset. The majority of videos came from YouTube as it has a larger international audience and has existed longer than YouTube Shorts.

4.1 Semantic information in video

Visual data can provide rich, semantically nuanced details of an event that are not captured in text documents due to reporting bias, limitations of text, and document length limits. To characterize the complexity of these videos and the information they provide, we annotate a set of two hundred videos of disasters in MultiVENT to identify visual entities in the videos that help answer common "who, what, where"-type questions about the events they depict.

We present videos of disaster footage to local annotators and provide them with a set of event-centric questions derived from FrameNet’s "disaster scenario" template [39]. We modify this template, designed to annotate the event semantics of text documents, to better cover the range of information provided by visual content. We instruct annotators to identify every on-screen entity (such as people, scrolling news headline banners, etc.) that might help answer one of these event-centric questions.

The template divides salient entities into six categories: The disaster itself ("what"), the location of the disaster ("where"), the time the disaster takes place ("when"), people affected by the disaster ("who") and first responders for the disaster, e.g., firefighters (also "who"), and any visible outcomes of the disaster. Not every category applies to both visual content and text: We exclude "where" and "when" from the set of categories that visual content should be annotated for (because identifiable depictions of "where" are present in almost every frame, and "when" in virtually none) and disaster outcomes from the set of text annotation categories, as textual examples of this category tend to involve full clauses, which complicate the annotation process.

We present the number of event-relevant entities that appear on-screen in these annotated videos in Table 1. For each annotated entity, we additionally ask annotators to rate their certainty that the entity is directly related to the event described by the video’s natural language description from 0% to 100%. We record these certainty scores in 20% intervals, i.e. as 20%, 40%, 60%, 80%, or 100%. The averages of the linguists’ confidence rankings by entity type are listed in Table 2.

As shown in Table 1, each video contains an average of 9.32 informative visual entities that pertain to the event in question. About half of these entities are purely visual, and half are within-video text that can be identified with an optical character recognition model. As indicated by Table 2, purely visual entities are more ambiguous than the text content shown onscreen alongside it, which aligns with past research that explores the difficulty humans have in interpreting visual content depicting complex events [41].

4.2 Video content by domain

As described in Section 3, we collect three main types of videos: Official news broadcasts, edited video footage, and raw, unedited footage. Of the 210 videos in the annotation set reported in Table 1, 53% are news broadcasts, 11% are edited footage, and 36% are raw footage. To quantify the

Table 1: Mean number of visual entities and in-scene text references (written text displayed within a video) present per video in a subset of 210 disaster videos from the current events dataset. We omit "where" and "when" entities from the visual content counts as "where" visual content technically appears in every frame and there are few types of visual evidence for "when" questions. We omit "outcomes" from the text references as an outcome by itself is a full event that is difficult to localize in text (this field is omitted from the FrameNet event template analogue for text documents).

	Visual entities	Text references	Total
Disaster ("What")	1.25	1.37	2.62
Place of occurrence ("Where")	-	1.54	1.54
Time of occurrence ("When")	-	0.77	0.77
Affected people ("Who")	1.22	0.54	1.76
First responders ("Who")	1.13	0.50	1.63
Disaster outcomes	1.00	-	1.00
Total	4.60	4.72	9.32

Table 2: Mean annotator certainty scores partitioned on entity type based on the annotations used for Table 1. 0.20 certainty indicates that the annotator is 20% sure that the annotated entity helps answer the tagged question about the described event, while 1.00 certainty indicates that the annotator is completely sure that the entity helps answer the tagged question about the event.

	Disaster	Where	When	AP	FR	Outcomes	All
Visual content	.787	-	-	.716	.765	.798	.830
Text content	.931	.907	.929	.856	.836	-	.900
Average	.862	.907	.929	.759	.787	.798	.865

difference in information presented by these different video types, we take the video annotations shown in Table 1 and partition these annotations by video type. We present the results in Table 3.

As shown by the results, news broadcasts depict the most relevant semantic information, followed by edited footage. This is particularly apparent when considering text content alone. On average, news coverage contains almost 9 times as much relevant on-screen text content than raw footage, and over three times more than edited footage. Visual content differences were less drastic, but news content still had two times more visual content than raw footage and 1.3 times more than edited footage. The difference in visual content between news coverage and edited footage is possibly due to average video length and the quality of the video curation — oftentimes, unprofessionally edited footage only draws from one source whereas news coverage draws from many.

4.3 Information evolution

As shown in Table 3, first-person footage is often opaque compared to professional coverage. However, comprehensive coverage often builds on earlier, less informative coverage. This can be seen in news cycles for slowly unfolding events and for sudden, unexpected events that take time to assess. This is illustrated in Figure 3, which shows a snapshot of the 2019 Notre Dame fire news cycle and demonstrates how unedited and poorly curated footage, often first-person witness accounts on social media, can be instrumental in the construction of our collective understanding of events. So, we propose that teaching models to understand different video formats, despite clear discrepancies in the amount of information they present, is important for developing robust systems.

5 Experiments

5.1 Approach

We consider the problem of teaching a model to map multilingual, natural language queries to multilingual video clips. Specifically, we consider a video set V and query set T with a indicator mapping function f that returns whether a query $t \in T$ describes a video $v \in V$. The model h is